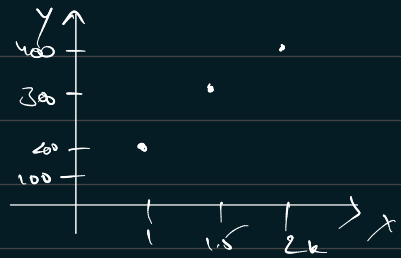


* Linear regression

Size (x)	Price (y)
1000	200
1500	300
2000	400



$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

$$J(\theta_0, \theta_1) = \frac{1}{2} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$(x^{(1)}, y^{(1)}) = (1000, 200)$$

$$(x^{(2)}, y^{(2)}) = (1500, 300)$$

$$(x^{(3)}, y^{(3)}) = (2000, 400)$$

bias term $x_0 = 1$

$$x^{(i)} = \begin{bmatrix} 1 \\ x^{(i)} \end{bmatrix} \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}$$

$$h_{\theta}(x^{(i)}) = (x^{(i)})^T \theta = \theta_0 \cdot (1) + \theta_1 (x^{(i)})$$

* Batch gradient descent

$$\frac{\partial J(\theta_0, \theta_1)}{\partial \theta_0} = \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot (1)$$

$$\frac{\partial J(\theta_0, \theta_1)}{\partial \theta_1} = \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

$x_0^{(1)}$	1	①
$x_1^{(1)}$	1	②

x_0	
1000	$\rightarrow 200$
1500	$\rightarrow 300$

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1$$

↓
only 1 feature

$$\theta = [\theta_0 \quad \theta_1]$$

$$x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ x_1 \end{bmatrix}$$

$$x^{(1)} = \begin{bmatrix} x_0^{(1)} \\ x_1^{(1)} \end{bmatrix} = \begin{bmatrix} 1 \\ 1000 \end{bmatrix}$$

$$x^{(2)} = \begin{bmatrix} x_0^{(2)} \\ x_1^{(2)} \end{bmatrix} = \begin{bmatrix} 1 \\ 1500 \end{bmatrix}$$

$$y = \begin{bmatrix} y^{(1)} \\ y^{(2)} \end{bmatrix} = \begin{bmatrix} 200 \\ 300 \end{bmatrix}$$

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

$$= \frac{1}{n} \left[\left(h_{\theta}(x^{(1)}) - y^{(1)} \right)^2 + \left(h_{\theta}(x^{(2)}) - y^{(2)} \right)^2 \right]$$

$$h_{\theta}(x^{(1)}) = 1 \cdot \theta_0 + \theta_1 (1000)$$

$$h_{\theta}(x^{(2)}) = 1 \cdot \theta_0 + \theta_1 (1500)$$

Q.2

Let $\theta_0 = 1$ & $\theta_1 = 0.1$

$$\theta = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}$$

$$\lambda = 0.01$$

1st iteration:-

$i=1$ (1st example):-

$$h_{\theta}(x^{(1)}) = \theta_0 + \theta_1 (1000)$$

$$= 1 + 0.1 (1000) = 101$$

$$i=2$$

$$h_0(x^{(2)}) = \theta_0 + \theta_1(1500) \\ = 1 + 0.1(1500) = 151$$

b) Calculate error:-

$$i=1 \\ \text{error}^{(1)} = h_0(x^{(1)}) - y^{(1)} \\ = 101 - 200 = -99$$

$$i=2$$

$$\text{error}^{(2)} = h_0(x^{(2)}) - y^{(2)} \\ = 151 - 300 = -149$$

c) Calculate gradient of cost function

$$\frac{\partial J(\theta_0, \theta_1)}{\partial \theta_0} = \frac{1}{n} \sum_{i=1}^n (h_0(x^{(i)}) - y^{(i)}) \cdot x_0^{(i)}$$

$$\frac{\partial J(\theta_0, \theta_1)}{\partial \theta_0} = \frac{1}{2} [(h_0(x^{(1)}) - y^{(1)}) \cdot 1 + (h_0(x^{(2)}) - y^{(2)}) \cdot 1]$$

$$\frac{\partial J(\theta_0, \theta_1)}{\partial \theta_0} = \frac{1}{2} [(-99) + (-149)] = \frac{1}{2} [-248] = -124$$

$$\frac{\partial J(\theta_0, \theta_1)}{\partial \theta_1} = \frac{1}{n} \sum_{i=1}^n (h_0(x^{(i)}) - y^{(i)}) \cdot x_1^{(i)}$$

$$= \frac{1}{2} [(h_0(x^{(1)}) - y^{(1)}) \cdot x_1^{(1)} + (h_0(x^{(2)}) - y^{(2)}) \cdot x_1^{(2)}]$$

$$= \frac{1}{2} [(-99) \cdot 1000 + (-149) \cdot 1500]$$

$$= -166250$$

Gradient vector

$$\nabla_{\theta} J(\theta) \approx \begin{bmatrix} -124 \\ -161250 \end{bmatrix}$$

d) Update the parameter vector

$$\begin{aligned}\theta_0 &:= \theta_0 - \alpha \frac{\partial J(\theta_0, \theta_1)}{\partial \theta_0} \\ &= 1 - 0.0001 \cdot (-124) \\ &= 1.00124\end{aligned}$$

$$\begin{aligned}\theta_1 &:= \theta_1 - \alpha \frac{\partial J(\theta_0, \theta_1)}{\partial \theta_1} \\ &= 0.1 - 0.0001 \cdot (-161250) \\ &= 1.7125\end{aligned}$$

Updated parameter vector

$$\theta = \begin{bmatrix} 1.00124 \\ 1.7125 \end{bmatrix}$$

e) Calculate cost (using updated θ_0 & θ_1)

$$\begin{aligned}h_0(x^{(1)}) &= 1.00124 + 1.7125 \cdot (1000) \\ &= 1713.50124\end{aligned}$$

$$\begin{aligned}h_0(x^{(2)}) &= 1.00124 + 1.7125(1800) \\ &= 2569.75124\end{aligned}$$

$$\begin{aligned}J(\theta_0, \theta_1) &= \frac{1}{n} \left[(1713.50 - 200)^2 + (2569.75 - 300)^2 \right] \\ &\approx 1860616.95\end{aligned}$$

⇒ The cost of the 1st iteration has increased from initial cost

$$\text{Iteration 2 (also c) with } \theta = \begin{bmatrix} 1.00124 \\ 1.7125 \end{bmatrix}$$

* Stochastic gradient descent :-

	dummy x_0	Size (x_1)	Price (y)
$x^{(1)} \rightarrow$	1	1000	200
$x^{(2)} \rightarrow$	1	1500	300

$$h_\theta(x) = \theta_0 + \theta_1 x_1$$

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}$$

$$y = \begin{bmatrix} 200 \\ 300 \end{bmatrix}$$

$$x^{(1)} = \begin{bmatrix} x_0^{(1)} \\ x_1^{(1)} \end{bmatrix} = \begin{bmatrix} 1 \\ 1000 \end{bmatrix}$$

$$x^{(2)} = \begin{bmatrix} x_0^{(2)} \\ x_1^{(2)} \end{bmatrix} = \begin{bmatrix} 1 \\ 1500 \end{bmatrix}$$

$$\text{Assume } \theta_0 = 1 \quad \& \quad \theta_1 = 0.1$$

$$\alpha = 0.00001$$

$$\theta = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}$$

SGD :-

Step 1 :- ($i=1$)

$$\begin{aligned} a) h_\theta(x^{(1)}) &= \theta_0 + x_1^{(1)} \theta_1 \\ &= 1 + 1000(0.1) \\ &= 101 \end{aligned}$$

b) Error :-

$$\begin{aligned} \text{error}^{(1)} &= (h_\theta(x^{(1)}) - y^{(1)}) \\ &= 101 - 200 = -99 \end{aligned}$$

c) Gradient of cost function -

$$J^{(i)}(\theta) = \frac{1}{2} [h_{\theta}(x^{(i)}) - y^{(i)}]^2$$

$$\frac{\partial J^{(i)}(\theta)}{\partial \theta_j} = (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x_j^{(i)}$$

for first example:-

→ for θ_0

$$\frac{\partial J^{(1)}(\theta_0, \theta_1)}{\partial \theta_0} = [h_{\theta}(x^{(1)}) - y^{(1)}] \cdot x_0^{(1)}$$

$$= -99$$

→ for θ_1 ($j=1$, $x_1^{(1)} = 1000$)

$$\frac{\partial J^{(1)}(\theta_0, \theta_1)}{\partial \theta_1} = [h_{\theta}(x^{(1)}) - y^{(1)}] \cdot x_1^{(1)} = (-99) \cdot 1000$$

$$= -99000$$

Gradient based on the first example is $\begin{bmatrix} -99 \\ -99000 \end{bmatrix}$

d) update parameter vector

$$\theta_0 := \theta_0 - \alpha \frac{\partial J^{(1)}(\theta_0, \theta_1)}{\partial \theta_0}$$

$$= 1 - (0.0001) (-99)$$

$$= 1.00899$$

$$\theta_1 := \theta_1 - \alpha \frac{\partial J^{(1)}(\theta_0, \theta_1)}{\partial \theta_1}$$

$$= 1 - (0.0001) (-99000)$$

$$= 1.09$$

